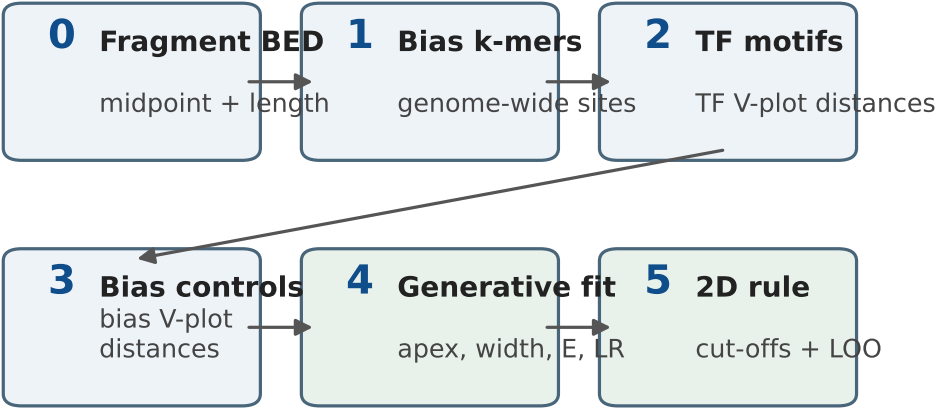
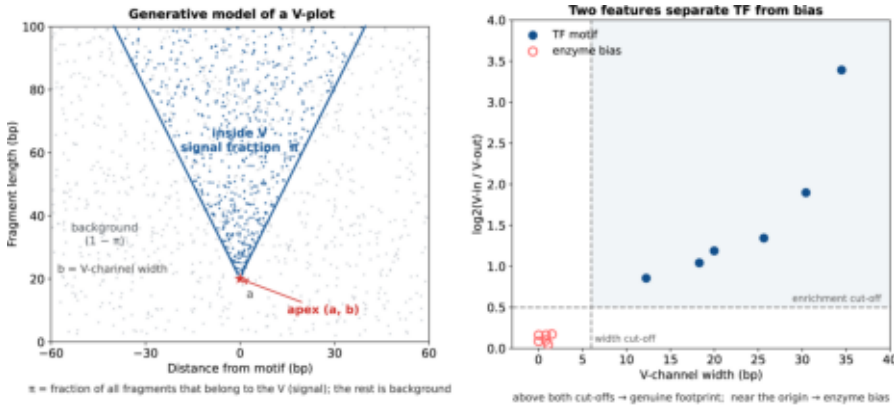


Portable Method B pipeline for V-plot footprint calling

A End-to-end workflow

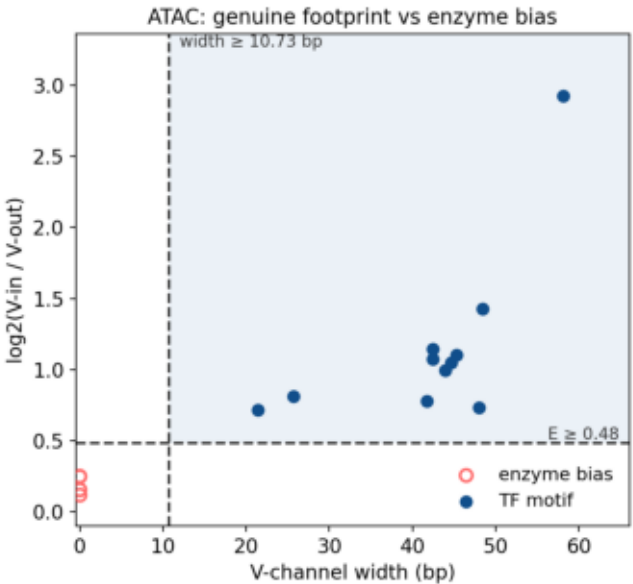


B Model-derived features



Output: apex-score TSVs, a width x enrichment scatter plot, and assay-calibrated cut-offs.

C Example calibrated scatter



D Example output: ATAC cut-offs and LOO validation

Feature	Cut-off	LOO acc	Sens	Spec
V-channel width	10.73 bp	1.00	1.00	1.00
Inside-V enrichment	0.48	1.00	1.00	1.00
2D rule	width ≥ 10.728 E ≥ 0.483	1.00	1.00	1.00

Cut-offs are learned from the gap between TF motifs and matched enzyme-bias controls. A motif is called a footprint only when both features exceed their cut-offs.